INDIAN INSTITUTE OF TECHNOLOGY MADRAS

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Roll No. | I | D | 2 | 5 | 5 | 0 | 2 | 7 |

Name : PUNEET

Total No. of Pages

Quiz I ☐ Quiz II/ Mid-Sem ☑ End-Semester ☐ Make-up ☐ Date : _____

Semester & Degree : II, MS Course No. MA5104 Part :

Question No.	1	2	3	4	5	6	7	8	9	10
Marks	15	5	30	26						

11	12	13	14	15	16	17	18	19	20	Total
										76

Answer on both sides of the paper including the space below

4.

(a) Observed Values

	O	NO	
MT	172	78	250
MSc	98	152	250
	270	230	500

$$E_{ij} = \frac{(\text{row total})\,(\text{column total})}{\text{total}}$$

$$= \frac{250 \times 270}{500} = 175$$ ✗

$$= \frac{250 \times 230}{500} = 115$$

expected values

	O	NO
MT	175	115
MSc	175	115

H_0 : programme & placement outcome are independent.

H_1 : programme & placement outcome are associated.

$$\chi^2 = \frac{\Sigma (O_{ij} - E_{ij})^2}{E_{ij}}$$

$$= \frac{(172-175)^2}{175} + \frac{(98-175)^2}{175} +$$

$$\frac{(152-115)^2}{175} + \frac{(78-115)^2}{175}$$

$$= \frac{(-3)^2}{175} + \frac{(-77)^2}{175} + \frac{(37)^2}{175} + \frac{(-37)^2}{175}$$

$$= \frac{9}{175} + \frac{5929}{175} + \frac{1369}{175} + \frac{1369}{175} = \frac{8676}{175}$$

$$\chi^2_{calc} = 49.57714$$

$$\nu = (r-1)(c-1)$$

$$= (2-1)(2-1) = 1 \times 1 = 1$$

11

$$\alpha = 0.05$$

$$\chi^2_{0.05,1} = 3.841$$

since, $\chi^2_{calc} > \chi^2_{\alpha,\nu}$ $(49.57714 > 3.841)$

therefore, reject H_0.

This proves that means at this confidence level we do not have enough evidence that programme and placement outcome are independent.

(b) Analytics profile →

Observed ⇒

	O	NO	
MT	160	40	200
MSc	42	8	50
	204	48	250

expected ⇒

	O	NO
MT	163.2	38.4
MSc	40.8	9.6

$= \frac{(\text{row total})(\text{col total})}{\text{total}}$

$= \frac{200 \times 204}{250}$

$= 163.2$

Also,

$= \frac{50 \times 204}{250} = 40.8$

Also,

$= \frac{200 \times 48}{250} = 38.4$

Also,

$= \frac{50 \times 48}{250} = 9.6$

$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$

$= \frac{(160-163.2)^2}{163.2} + \frac{(40-38.4)^2}{38.4} + \frac{(42-40.8)^2}{40.8} + \frac{(8-9.6)^2}{9.6}$

$= \frac{10.24}{163.2} + \frac{2.56}{38.4} + \frac{1.44}{40.8} + \frac{2.56}{9.6}$

$= 0.0627 + 0.0666 + 0.0352 + 0.2666$

$= 0.4311$

H_0: under analytics, these 2 are indep.

H_1: " " , these are associated

$\chi^2_{calc} < \chi^2_{0.05,1}$

$\nu = (r-1)(c-1) = (2-1)(2-1) = 1$

$0.4311 < 3.841$

fail to reject H_0

means under analytics, they are independent.

quantitative finance ⇒

Observed →

	O	NO	
MT	12	38	50
MSc	56	144	200
	68	182	250

Expected values ⇒

$\frac{50 \times 68}{250} = 13.6$

$\frac{200 \times 68}{250} = 54.4$

$\frac{50 \times 182}{250} = 36.4$

$\frac{200 \times 182}{250} = 145.6$

	O	NO
MT	13.6	36.4
MSc	54.4	145.6

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

~~$= \frac{(12 - 13.6)^2}{13.6} + \ldots$~~

~~$= \frac{(13.6 - 12)}{}$~~

$$= \frac{(12 - 13.6)^2}{13.6} + \frac{(56 - 54.4)^2}{54.4} + \frac{(38 - 36.4)^2}{36.4} + \frac{(144 - 145.6)^2}{145.6}$$

$$= 0.1882 + 0.0470 + 0.0703 + 0.0175$$

$$= 0.323$$

$$\nu = (r-1)(c-1)$$
$$= (2-1)(2-1)$$
$$= 1 \times 1 = 1$$

$0.323 < 3.841$

fail to reject H_0

means under quantitative profiles, they are independent.

(15)

(c) Yes they both point towards opp. directions
Yes this is possible, when we had both profiles, they were associated. But individually, they are independent of each other.

(0)

The placement committee must think some where things are going wrong or may be broadly things are associated but internally they aren't.

3.

(i) $n_1 = 14, \quad \bar{x}_1 = 17, \quad s_1^2 = 1.5$

$n_2 = 16, \quad \bar{x}_2 = 19, \quad s_2^2 = 1.8$

$\alpha = 0.01$

$H_0 : \mu_1 = \mu_2$
$H_1 : \mu_1 \neq \mu_2$
} null & alternate hypo

2 popula^n are normal with equal variances,

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(13)(1.5) + (15)(1.8)}{30 - 2}$$

$$= \frac{19.5 + 27}{28} = 1.6607$$

Parameter of interest is $(\mu_1 - \mu_2)$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$s_p = \sqrt{1.6607}$
$= 1.2887$

$\sqrt{\frac{1}{14} + \frac{1}{16}}$
$= \sqrt{\frac{16+14}{224}} = \frac{\sqrt{30}}{\sqrt{224}}$
$= 0.3659$

$$= \frac{-2}{(1.2887)(0.3659)}$$

$$= \frac{-2}{0.4715} = -4.241$$

$$\nu = n_1 + n_2 - 2$$
$$= 16 + 14 - 2 = 28$$

$\alpha = 0.01$

$\alpha/2 = 0.005$

$t_{0.005, 28} = 2.763$

$|t_{calc}| > t_{\alpha/2, \nu}$ $\Rightarrow$ reject H_0.

$|-4.24|$ $> t_{0.005, 28}$ $\rightarrow$

$\therefore$, reject H_0

means the two medications differ in mean recovery time

(18)

(ii) $(\mu_1 - \mu_2) \pm (t_{0.01, 28})(SE)$ $\alpha = 0.01$

$SE = 0.4715$ (from i)

$-2 \pm (2.467)(0.4715)$

1.16319

-2 ± 1.16319

~~$(-3.16319, -0.83681)$~~

$(-3.16319, -0.83681)$

CI

Now, since

~~0 ∈ CI, then~~

0 ∉ CI, then again reject H_0

yes they must agree irrespective of the method used why? (b)

(iii) $H_0: \mu \geq \mu_2$

$H_1: \mu_1 < \mu_2$

(b)

$$t = \frac{\mu_1 - \mu_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= -4.241$$

$|-4.241| > 2.763$ ✗

reject H_0

that means $\mu_1 < \mu_2$

so, test statistic didn't change. Though some of our signs did change in the ~~my~~ null hypothesis the decision is $\mu_1 < \mu_2$

2.

(a) type II ⇒ accept the shipment

H_0 : accept the shipment

H_0 : do not accept the shipment / return to buyer

(b) ~~α type~~ α = type I error

(reject H_0 | H_0 is true)

~~was rejected~~ even though shipment should have been accepted, we ejected it. Means supplier bears this risk.

1.

(i) C ✓

(ii) C ✗

(iii) C ✓

(iv) B ✓

(v) B ✗

(vi) A ✓

(vii) A ✓

(viii) B ✓

15

— end —